

GESC + Robust Gaussian V1 Final Project Report

Architecture, implementation, simulation and physical evidence, limitations, and reproducibility

DSIM Lab

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V1 evidence boundary. This report preserves selected successes, failed broad-readiness gates, partial and withheld results, and the incomplete second-extremum physical boundary. It does not claim broad simulation or physical robustness and does not describe an implemented V2.

Canonical reader-facing artifact: `FINAL_PROJECT_REPORT_V1.pdf`

Authoritative source: `FINAL_PROJECT_REPORT_V1.tex`

Repository audit companion: `FINAL_PROJECT_REPORT_V1.md`

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1. Document identity and V1 evidence boundary

This is the repository-native content and audit companion for the canonical typeset **V1 report PDF**. Its authoritative typesetting source is [FINAL_PROJECT_REPORT_V1.tex](#). The three synchronized representations form the thesis-facing closeout report for the DSIM_GESC_Gaussian_Codex_Implementation_Package effort on branch feature/gesc-gaussian-robustness-v1. It was prepared on 2026-08-12 against parent/base HEAD e3dd0ef5cfa5a79ee8bed3017bf844f4f269add7. After the documentation gates passed, the user authorized one closeout commit with the exact subject `docs(phase10): close GESC Gaussian V1 report and evidence`. Because a commit cannot embed its own object ID without changing that ID, the authoritative V1 boundary is the commit containing this report set with that exact subject; resolve it with `git log -1 --format=%H -- docs/codex/gesc_gaussian/FINAL_PROJECT_REPORT_V1.pdf`. The future V2 branch must start from that containing commit, not from e3dd0ef alone.

This report covers the repository implementation, package requirements, source material, Phases 00-10, simulation evidence through terminal v8.12, physical snapshot integration and eight retained attempts, tests, artifacts, limitations, and reproducibility boundaries. Its machine-checkable companion is [the Phase 10 coverage matrix](#). The exact validation commands and outcomes are in [the Phase 10 documentation validation](#).

V1 is a frozen evidence record. It may be cited, reproduced within its stated scope, or used as a baseline, but its failed and partial results must not be rewritten when later work improves the algorithm. The intended future branch name is feature/gesc-gaussian-robustness-v2; no V2 branch, implementation, or result exists in this report.

Claim vocabulary

Term	Meaning in this report
Implemented	Present in the cited current source owner. It does not imply empirical success.
Software gate passed	The cited build, focused test, static, or infrastructure criterion passed.
Behavioral pass	The predeclared behavioral observations for that one run or scope passed.
Formal pass	Every selected predicate in that exact versioned contract passed.

Term	Meaning in this report
Selected demonstration	Evidence for a named fixed layout/profile; not an unbiased robustness population.
Partial	Useful behavior or infrastructure occurred, but the complete declared result did not pass.
Failed	A declared gate missed. The result remains evidence and is not relabeled.
Withheld/not run	Execution stopped under a declared early-stop rule; no outcome is inferred.
Broad readiness	The entire predeclared simulation- or physical-ready gate passed. V1 achieved neither.

The source-of-truth order used here is current source/tests/resolved interfaces and launch graph; Git state/history; completed handoffs; approved Plans; live status/checkpoints; Phase 00 audits and the Phase 00-05 bridge; package specifications; then raw transcripts or experimental memory. Historical documents describe their date-specific state. A later dated amendment may supersede a stale “current” sentence without altering the original file.

2. Executive summary and final conclusions

V1 built a complete, observable, opt-in GESG + Gaussian research stack while preserving legacy ESC behavior. The stack includes six robust typed messages, an explicit eight-state supervisor, synchronized basin estimation, adaptive anisotropic Gaussian fills, revision-aware merging, pure and assisted escape, bounded recentering, a sole rosbag recorder and validator, deterministic scenario execution, offline analysis, simulation validation contracts, and a selected physical two-light workflow using the same algorithm implementation.

The strongest supported simulation results are narrow and useful:

- v8.10 primary fixed two-source 1:4 layout: 11/11 formal passes;
- v8.11 secondary fixed two-source 1:4 layout: 6/6 formal passes;
- v8.12 visible interior-anchor probe: all 14/14 selected predicates passed;
- the first varied v8.12 matrix case completed the intended science but passed 13/14 formal predicates because exit alignment was 0.735563, below the frozen 0.80 threshold; the other three matrix cases were withheld.

Those results did **not** pass the broad Phase 08 robustness gate. Earlier v1, v2, v3/V3A-D, V4, V5, V6, Phase 08.7, and v8-v8.9 failures remain material: they show activation gaps, route/topology errors, evidence defects, recovery nonrepeatability, and sensitivity across starts/seeds. No simulation-ready tag exists.

The strongest physical result is the eighth retained run. It completed local convergence and nine-second classification, created one Gaussian fill, performed repulsive plus assisted escape, returned to SEARCH, and reacquired a stronger-light signal. Onboard /odom integrated 7.797691 m; five later rotation-scale voltage peaks had median 1.9013 V, 4.37 times the local candidate estimate of about 0.435 V. The operator stopped before a second convergence confirmation or GOAL_HOLD. Completeness was 61/62; the only failure was a retained cross-topic shutdown-ordering false negative involving one in-flight rotation status. This is first physical two-basin behavioral success, **not** complete second-extremum acceptance and not broad physical readiness.

The decision-useful V1 conclusion is therefore:

The project demonstrated that one shared GESG + adaptive Gaussian pipeline can detect and fill an undesired light basin, escape it, and progress toward a stronger source in selected simulation layouts and one retained physical trial. V1 did not demonstrate robustness over the declared broad simulation envelope, arbitrary layouts/intensities, or a complete second-extremum physical run.

The appropriate next step is a separately planned V2 study with new versioned evidence, not retroactive tuning or gate changes in V1.

3. Research motivation and supplied source material

The project began with a GESC controller plus Gaussian local-minimum escape whose physical demonstration could orbit repeatedly, leave a basin slowly, or create residual minima. Dr. Nili's direct requirements were to focus on GESC + Gaussian robustness, treat raw cost, Gaussian repulsion, and affine assistance as independently switchable, test extensively in Gazebo, cover different depths/strengths/source counts/close minima, and avoid presenting many repeated circles as success. Light remained the implemented field; reflective acoustic fields were future motivation only.

Patrick's data discussion required the input and output of each software block, switching times, modes, Gaussian parameters, commands, and timestamps to be published as ROS data and retained in rosbag2. Console text was explicitly not enough. His whiteboards motivated—but did not mandate—the basin-scale fill, soft association, and switchable cost interpretation.

The supplied material is indexed in [the source-material README](#) and includes:

- meeting_dr_nili_audio.m4a, an automatic extracted transcript, and the raw transcript document;
- meeting_patrick_data_collection_raw.txt;
- [the Gaussian-escape whiteboard](#);
- [the switchable-cost/recenter whiteboard](#); and
- [consolidated meeting decisions](#).

Automatic transcripts may contain recognition errors. This report uses the consolidated decisions for attribution and labels kernel estimation, quadratic fitting, fill escalation, exact state sequencing, and recenter mechanics as adopted engineering hypotheses rather than direct quotations or immutable meeting mandates.

4. Requirements, assumptions, and decision traceability

The fixed invariants were: preserve the raw minimization sign and units; autonomous escape with bounded safe motion; avoid known minima; classify the goal using evidence unavailable to neither simulator nor physical controller; record complete structured data; switch cost components independently; preserve selectable legacy behavior; and keep simulation/physical algorithm semantics shared.

The implemented cost contract is

$$J_{\text{aug}}(x, t) = w_s(t)J_{\text{raw}}(x) + w_g(t) \sum_i F_i(x) + w_a(t)J_{\text{affine}}(x).$$

The physical adapter records sensor voltage +V and exposes minimization cost -V; the simulator retains its model-relative cost. The sensor continues to be sampled and logged while $w_s=0$. Ground-truth source geometry, declared roles, Vicon, and evaluator topology never enter the controller.

Policy and amendments

Boundary	V1 treatment
Level A contradiction	Stop for objective, sign/unit, ownership, parity, safety, evidence-integrity, physical-authorization, or overlapping-user-change conflicts.
Level B correction	Allow a bounded, tested correction that preserves the objective and gates; record the failed evidence and exact amendment.
Level C failure	Close that fixed experiment version honestly, retain every result, and require a separately planned next version.
Legacy behavior	algorithm_profile=legacy remains the direct-launch default; robust behavior is opt-in.

Boundary	V1 treatment
Broad Phase 08 gate	Failed and never converted into a selected-demonstration claim.
Phase 09 amendment	Allowed selected-scenario source integration and later explicitly authorized motion without a broad readiness tag.
V2	Future branch and new evidence only; V1 files and outcomes remain immutable.

The main bounded corrections included ROS launch DOUBLE typing and signal-safe shutdown (Phase 04), recorder/-parameter/readiness sequencing (Phases 05-08), real Gazebo contact/delay evidence (07.5), scenario/evaluator contracts and runner faults (Phase 08), and physical clock/startup/executor/rotation races (Phase 09). Each correction received new versioned evidence; none changed an already-observed failure into a pass.

5. Repository baseline and final architecture

Phase 00 found three canonical simulation packages:

- `ros_esc`: existing algorithm, cost, filter, controller, Gaussian, recording, scenario, validation, and analysis owners;
- `ros_esc_interfaces`: the only custom ROS interface package; and
- `turtlebot3_rotating_sensor`: Gazebo model, launch graph, worlds, and wrappers.

The physical source is a read-only snapshot at `/home/mattb/physical_TB3_files_snapshot/pi/ros2_ws/src`, containing `ros_esc`, `ros_esc_interfaces`, and `turtlebot3_vehicle_nodes`. Phase 09 extended the same algorithm owners and added only hardware/evaluation/launch adapters. The current local snapshot has 347 regular files; the retained M8L inventory has 435 rows including directories. Phase 10 did not access the live Pi filesystem.

Simulation graph

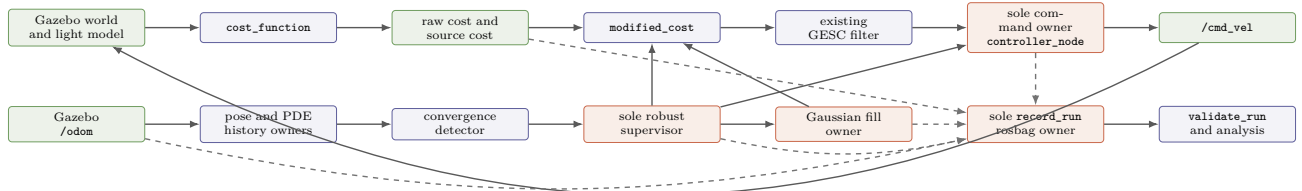


Figure 1: Canonical simulation ownership and evidence graph. Solid arrows are runtime/control flow; dashed arrows are recorder evidence inputs. The graph has one supervisor, one command owner, one recorder, and one validator path.

`gazebo.launch.xml` is the single central launch. Cost owners are mutually exclusive, the robust supervisor is the sole robust state owner, and `custom_controller` remains the sole `/cmd_vel` owner. The supervisor supplies weights, authorization, and bounded supervisory contribution; it does not create a second velocity publisher.

Selected physical graph

Wheel/IMU-backed `/odom` is the sole algorithm pose. Vicon publishes `nav_msgs/msg/Odometry` on `/gesc_gaussian/evaluation/vicon_odom` for passive evaluation and cannot gate or stop motion. The selected open field has no autonomous obstacle or wall avoidance. Arrival and run end are manual operator Ctrl+C; `GOAL_HOLD` is not a physical process terminator.

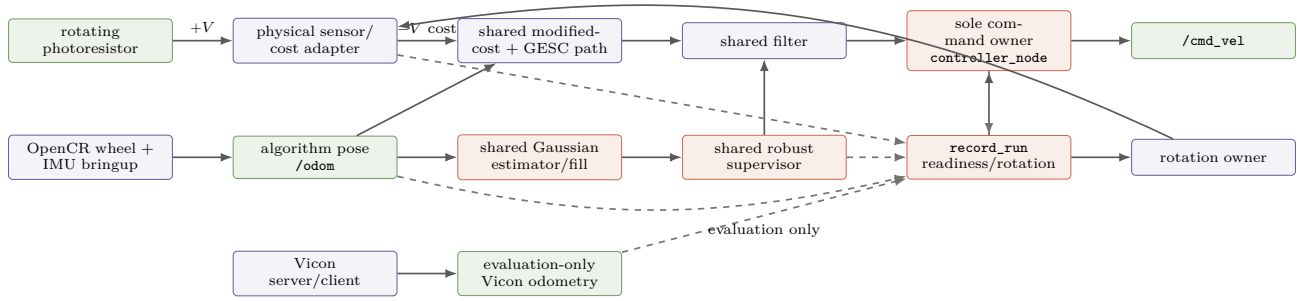


Figure 2: Selected physical graph. Wheel/IMU-backed /odom is the algorithm pose; Vicom is passive evaluation only. The recorder owns readiness, rotation authorization, and final-zero coordination.

Legacy and robust selection

Direct Gazebo and physical launch files default to legacy and disabled PDE/ robust extensions. The selected wrappers explicitly choose robust_gaussian_v1, PDE extensions, typed observability, and recording readiness. Legacy arrays, topics, wrappers, configs, and historical Heavy-Ball examples remain present. Heavy-Ball is archive/reproduction context, not the active GESc + Gaussian V1 method.

6. GESc and robust Gaussian mathematics

6.1 Cost and sign contract

The controller is formulated around minimization. For channel c , the observable cost pipeline is

$$J_{\text{aug},c}(x, t) = w_s(t)J_{\text{raw},c}(x, t) + w_g(t)J_{\text{fill},c}(x, t) + w_a(t)J_{\text{affine},c}(x, t).$$

modified_cost_node is the sole composer. It never changes the meaning of the raw term. In simulation, J_raw has the units and scale of the selected light model. On the physical robot, photoresistor_node records raw_sensor_value=+V and publishes raw_cost=-V; therefore a stronger light is a lower raw cost. source_score is a separate classification quantity and was deliberately invalid/NaN in the final physical profile because physical source-score calibration was inert. Raw voltage and -V cost remained valid.

The state-dependent weights are fixed by [state_machine.py](#):

State	w_s	w_g	w_a	Interpretation
SEARCH	1	1	0	Follow the raw field while retaining fills.
VERIFY_EXT REMUM	1	1	0	Hold the same objective while classifying a candidate.
DESIGN_OR_ MERGE_FILL	0	1	0	Remove raw attraction while the fill is made authoritative.
ESCAPE_REP ULSE	0	1	0	Leave under Gaussian repulsion.
ESCAPE_ASSIS T	0	1	1	Add the bounded affine escape direction.

State	w_s	w_g	w_a	Interpretation
RECENTER	0	1	0	Supervisor owns bounded return motion; raw attraction stays off.
GOAL_HOLD	0	1	0	Zero-motion terminal algorithm state.
FAILSAFE	0	0	0	Zero all objective components and command motion to zero.

The affine contribution implemented in [modified_cost_script.py](#) is

$$J_{\text{affine}}(x, t) = -s b(t)^{\top} (x - x_a), \quad b(t) = b_0 e^{-\lambda(t-t_a)},$$

where $s = \text{affine_direction_sign}$, x_a is the fill/approach anchor, and b_0 comes from PDE history or the odometry fallback. V1 selected $s = +1$, gain 0.50, decay $5e-7$, and maximum age 35 s. A stale/degenerate direction is invalid rather than silently reused.

6.2 GESC path

The existing GESC owners were extended, not replaced. The rotating sensor and pose owners feed `cost_function_node`/the physical adapter and `pde_history`; `modified_cost_node` emits the selected objective; `filter_node` performs the existing demodulation/filter dynamics; and `controller_node` converts the filtered estimate plus any supervisor contribution into the sole `/cmd_vel`. `GescDiagnostics` records filter input/output, state before/derivative/after, and dither phase/amplitude/frequency. `ControlDiagnostics` records unsaturated GESC, supervisor, combined, final, saturation, limit, and gain values.

This separation matters empirically. A roughly three-second rotating-sensor cycle can produce bounded phase-locked command ripple; it is not by itself an escape event. State and event messages, rather than visual motion alone, identify convergence, fill creation, escape, and goal classification.

6.3 Basin estimator

[gaussian_fill_script.py](#) synchronizes pose and source cost within `sample_sync_tolerance_sec`, rejects nonfinite, stale, implausibly fast, and MAD-outlier samples, and retains a bounded window. Let accepted samples be (x_k, J_k) , normalized costs be $\tilde{J}_k$, position bandwidth be h , and cost temperature be T_J . The stable joint spatial/cost kernel is

$$q_k = \exp\left(-\frac{\|x_k - \mu\|^2}{2h^2} - \frac{\tilde{J}_k}{T_J}\right), \quad \bar{q}_k = \frac{q_k}{\sum_j q_j},$$

$$\mu = \sum_k \bar{q}_k x_k, \quad \Sigma = \sum_k \bar{q}_k (x_k - \mu)(x_k - \mu)^{\top}.$$

Mean-shift iterations refine the center; covariance eigenvalues are clipped to the configured finite interval. The implementation also fits a regularized local quadratic

$$\hat{J}(x) = c + g^{\top}(x - \mu) + \frac{1}{2}(x - \mu)^{\top} H (x - \mu)$$

with ridge quadratic_ridge_lambda. Excess condition number, insufficient samples, shallow basin depth, nonpositive curvature, or invalid geometry lowers confidence or rejects the design. Candidate-informed mode carries the pretrigger raw-cost distribution into amplitude design; evaluator topology and ground truth are not inputs.

6.4 Adaptive fill design and validation

For fill i , V1 uses an anisotropic positive cost lift

$$F_i(x) = A_i \exp\left[-\frac{1}{2}(x - \mu_i)^\top \Sigma_i^{-1}(x - \mu_i)\right].$$

The covariance is scaled, then its principal widths are clipped between the sigma floor and ceiling. Amplitude combines measured basin depth and local curvature, then is clipped. Selected v8.12 used covariance scale 2.5, sigma range 0.50--1.25 m, depth scale 1.5, amplitude maximum 6.25 cost units, and candidate-informed scale 1.25.

Before publication, a finite grid over the support checks that the augmented surface no longer retains the prohibited local minimum. A failing design may increase amplitude by 1.5 and width by 1.25, up to five escalations; a still-invalid design produces FILL_DESIGN_FAILED, not a nominal fill. The reported support and exit geometry are

$$r_{\text{support}} = k_s \sigma_{\text{max}}, \quad r_{\text{exit}} = k_e \sigma_{\text{max}},$$

with selected $k_s=3.0$ and $k_e=2.70$. The supervisor requires radial exit and a one-second hold before declaring the basin left.

6.5 Association, merge, revision, and confidence

New candidates are compared with active fills using bandwidth/radius and a soft association probability. `minimum_merge_probability=0.60` is the soft merge gate; geometric hard-overlap protection prevents two active fills from silently representing the same basin. A merge creates a new revision and marks the previous representation superseded. Fill identity (`fill_id`), cluster identity, revision, active/superseded flags, fit residual, condition number, sample count, confidence, and escalation count are published in `GaussianFill`. This makes update history reconstructable from the bag.

The low-confidence threshold is 0.60. Low confidence is observable and can block unsafe assumptions; it is not converted into false precision. Numeric safeguards include finite checks, positive-duration and nonnegative-range validation, eigenvalue clipping, a 1e-9 grid-minimum tolerance, bounded sample buffers, explicit validity flags, and zero-output fallbacks.

7. Hybrid state machine and safety behavior

The supervisor owns one eight-state machine. The message constants and Python enum intentionally match.

State	Required behavior	Principal exit/timeout
SEARCH	Raw+fill objective, observe convergence, reset stale search epochs where selected.	Qualified candidate; stale/invalid input; external stop.
VERIFY_EXTREMUM	Hold candidate evidence for complete rotations; compare counted candidates or calibrated score.	Goal, undesired candidate, rejection, or 12 s verification timeout.
DESIGN_OR_MERGE_FILL	Freeze raw attraction, request one design/merge, await an attributable fill event.	Valid fill or 5 s design timeout/failure.
ESCAPE_REPULSE	Use active fill only; measure radius/progress and maintain controller ownership.	Exit+1 s hold, stall, or selected 35 s limit.

State	Required behavior	Principal exit/timeout
ESCAPE_ASSIST	Add bounded affine/supervisor-owned assist when selected and repulsion stalls.	Exit+hold or escape timeout.
RECENTER	Bounded supervisor motion to a safe target while avoiding fill support and walls.	Position+dwel or 30 s timeout. Disabled in selected v8.12/physical profile.
GOAL_HOLD	Publish zero-motion state after goal evidence.	Manual/external shutdown; it does not terminate recording.
FAILSAFE	Set weights and command to zero; publish attributable failure.	Explicit process shutdown/restart only.

Selected counted-candidate classification assumes exactly two sources and one fillable local candidate. It gathers three full 3 s rotations per candidate and retains six rotations of pretrigger evidence; median/MAD statistics rank the second candidate against the first. This is an experimental contract, not general unknown-source inference.

Safety is layered:

1. Startup requires the selected profile, compatible PDE/controller owners, fresh pose/sensor/state, and—on physical runs—recorder readiness.
2. Stale pose, sensor, state, command, recorder heartbeat, or rotation lease leads to a bounded zero/failsafe path.
3. Only controller_node publishes /cmd_vel; the supervisor contributes to that owner rather than publishing a competing command.
4. zero_command_on_shutdown=True and the managed recorder revoke readiness and rotation, publish stop, wait for stable base/rotation zero, then stop target and bag.
5. The selected physical environment is open field and has no autonomous obstacle avoidance. A human operator remains responsible for the e-stop, clearance, and manual Ctrl+C.

8. ROS 2 implementation reference

8.1 Runtime owners and entry points

Owner	Responsibility	Installed entry point(s)
cost_function_node / physical photoresistor_node	Raw sensor and raw-cost source; exactly one active cost owner.	cost_function_node; vehicle photoresistor_node
pde_history_node, pde_cost_history_node	Pose/cost history used by GESG and affine direction.	same names
convergence_detector_node	Qualified candidate and confirmation events.	convergence_detector_node
gaussian_fill_node	Sample synchronization, basin estimation, design, validation, registry.	gaussian_fill_node
modified_cost_node	Weighted raw+Gaussian+affine composition.	modified_cost_node
filter_node	Existing GESG demodulation/filter dynamics.	filter_node
supervisor_node	Sole robust state, weights, fill requests, escape/recenter policy.	supervisor_node
controller_node	Sole final command and /cmd_vel publisher.	controller_node
record_run / validate_run	Sole managed recorder and sole completeness validator.	record_run, validate_run

Owner	Responsibility	Installed entry point(s)
run_scenario / validate_robustness	Simulation-only deterministic orchestration and gate evaluation.	run_scenario, validate_robustness
bag analysis	Synchronized tables, CSVs, plots, run/matrix summaries.	analyze_run, summarize_matrix
physical base/Vicon/rotation adapters	No-lidar base /odom, passive Vicon odometry, recorder-authorized sensor rotation.	vehicle bringup, odometry_node, phase09_rotation_node

The canonical simulation launch is [gazebo.launch.xml](#). It has compatibility defaults `algorithm_profile=legacy`, `use_pde_extensions=False`, and `enable_observability=False`; a scenario or wrapper must opt into `robust_gaussian_v1`. The selected physical launch is `turtlebot3_vehicle_nodes/launch/gesc_gaussian_two_source.launch.xml` in the read-only snapshot, called by `gesc_gaussian_two_source_voltage.bash`. Direct physical defaults are likewise `legacy/inert`; the wrapper supplies the robust profile.

8.2 Typed messages

Six robust messages were added; five pre-existing stamped/timekeeper messages remain byte-compatible.

Message	Complete semantic payload
AlgorithmState	Current/previous enum and names, transition reason, run/profile, elapsed time, active fill/escape geometry, radial progress, stall, safe direction, recenter target, weights, failsafe, validity/source timestamps.
AlgorithmEvent	Typed configuration/capability/state/convergence/fill/-goal/escape/recenter/timeout/failsafe events, state, fill ID, reason/detail, named numeric values, validity/source timestamps.
GaussianFill	Frame, fill/cluster/revision, center, amplitude, covariance, principal widths/orientation, support/exit radii, confidence, samples, residual/condition/escalations, active/superseded and validity.
CostBreakdown	Source mode/name/channels; raw and filtered sensor; raw cost; source score; Gaussian, affine, and augmented costs; three weights and validity flags.
GescDiagnostics	Filter input/output/state derivative path plus dither phase, amplitude, frequency and validity.
ControlDiagnostics	GESG, supervisor, combined, and final six-axis commands; saturation, limits, <code>k_vx</code> , <code>k_wz</code> , validity.
Legacy set	StampedFloat64, StampedFloat64MultiArray, StampedString, StampedTransformMultiArray, and Timekeeper remain available for old consumers.

8.3 Canonical topics and role boundaries

Topic	Type/role	Owner/boundary
/odom	nav_msgs/Odometry; algorithm pose	Gazebo/base only; sole physical control pose.
/cmd_vel	final base command	controller_node only.
/gesc_gaussian/source_cost	algorithm source-cost stream	active simulation/physical cost owner.

Topic	Type/role	Owner/boundary
/gesc_gaussian/cost_breakdown	CostBreakdown	cost/modified-cost observability.
/gesc_gaussian/gesc_diagnostics	GescDiagnostics	filter owner.
/gesc_gaussian/control_diagnostics	ControlDiagnostics	controller owner.
/gesc_gaussian/convergence_status	legacy-compatible convergence status	detector owner.
/gesc_gaussian/fill_requests	attributable fill request	supervisor to fill owner.
/gesc_gaussian/gaussian_fills	GaussianFill	fill registry owner.
/gesc_gaussian/algorithm_state	AlgorithmState	supervisor owner.
/gesc_gaussian/algorithm_events	AlgorithmEvent	attributable publishers; state semantics owned by supervisor.
/gesc_gaussian/supervisor_command	bounded contribution	supervisor to sole controller.
/gesc_gaussian/recording_ready	managed-run heartbeat	record_run.
/gesc_gaussian/stop_requested	managed stop	record_run/operator shutdown path.
/gesc_gaussian/rotation_authorized	physical rotation lease	record_run only.
/gesc_gaussian/evaluation/vicon_odom	nav_msgs/Odometry; passive evaluation	odometry_node; never a controller input or motion gate.
simulation delayed/contact topics	disturbance/evaluation support	scenario support only; default off.

QoS details that affect evidence are installed with the recorder manifest and [qos_overrides.yaml](#). The bag is the authority: console logs and derived CSVs cannot replace a required typed topic.

9. Complete parameter reference

This section distinguishes node defaults, direct-launch compatibility defaults, and selected experiment overrides. Numeric constraints are enforced in their cited owner; durations and scales are finite and normally positive, counts are integer/nonnegative or positive as stated, probabilities are in $[0,1]$, covariance/sigma values are positive, and topic/profile strings must resolve. The exhaustive launch forwarding surface remains in the canonical XML; the tables below cover every parameter that changes the documented robust algorithm, evidence, or selected physical behavior.

9.1 Convergence defaults and selected V1 values

Parameter	Node/direct default	Selected v8.12 and physical	Unit/constraint
k_periods	20	inherited	periods; positive int
threshold / launch	0.1 / 0.2	launch-selected model value	cost/filter scale;
convergence_threshold			positive
decay_rate	0.15	inherited	positive
n_buffer, omega	2000, 5.0	inherited	samples, rad/s;
			positive
min_fill_periods	1.0 / launch 2.0	2.0	periods; positive
convergence_count_start	3	inherited	positive int
reset_counter_after_event	true	true	bool
convergence_confirmation_policy	crossing_count	qualified_dwell	enum
convergence_confirmation_dwell_sec	6.0	6.0	s; positive

Parameter	Node/direct default	Selected v8.12 and physical	Unit/constraint
convergence_confirmation_exit_threshold_scale	1.5	1.5	positive
state_gating_enabled	false	true	bool
minimum_path_length_m	0.0	0.20	m; nonnegative
maximum_path_efficiency	1.0	0.50	ratio; [0,1]

9.2 Gaussian estimator, designer, and registry

Parameter family	Current defaults	Selected override	Units/constraints
Legacy compatibility	escape_policy=conditional_gaussian_fill; amplitude=5.0; min_sigma=0.10; max_sigma=5.0; min_points=50; use_recent_fraction=1.0; max_fills=1; cooldown/min fill distances 0; center_source=event_mean; max fit-center distance 0.75; fit amplitude [0.15,10]; offset bound 10	max fills 1	Legacy path only unless forwarded; finite bounds.
Synchronization	pose /odom; channel 0; tolerance 0.05; maximum speed 0.20; outlier MAD 3.5; max samples 4000; window 8.0; minimum samples 40; max age 12.0	minimum samples 40	m, s, m/s, count; positive/nonnegative.
Center/covariance	mean-shift iterations 5; tolerance 0.005; kernel bandwidth 0.25; cost temperature 0.05; covariance eigenvalues [0.0025,0.25]	inherited	m, normalized cost, m ² ; ordered positive limits.
Quadratic/depth	ridge 1e-6; max condition 1e8; center/shoulder percentiles 10/80; inner Mahalanobis radius 1.0; minimum basin depth 0.02	inherited	finite; percentiles [0,100].
Width	covariance scale 2.5; sigma floor/ceiling 0.15/1.25	2.5; 0.50/1.25	scale, m; positive and ordered.
Amplitude	depth scale 1.5; curvature scale 1.2; min/max 0.10/3.00	depth 1.5; max 6.25; candidate scale 1.25	cost units; finite nonnegative.
Residual validation	grid 41; support sigma 3.0; max escalations 5; amplitude/width factors 1.5/1.25; tolerance 1e-9	inherited	positive odd grid/count/scales.

Parameter family	Current defaults	Selected override	Units/constraints
Geometry	support sigma 3.0; exit sigma 2.5	exit 2.70	sigma multipliers; positive.
Association	merge bandwidth 0.50; radius scale 2.0; minimum probability 0.60; low-confidence 0.60	inherited	m, scale, probabilities.
Redesign/candidate	retain samples false; candidate-informed false; scale 1.0	true; true; 1.25	bool/positive.

9.3 Supervisor and selected robust profile

Family	Direct/current default	Selected v8.12 and physical	Constraint/unit
Profile/rate/start	robust_gaussian_v1 node; direct launch legacy; publish 20 Hz; startup 5 s	robust; 20 Hz; physical startup 100 s	valid profile, positive Hz/s
Goal evidence	score 0.95; rotation 3 s $\times 2$; goal hold 3 s	same	calibrated ratio or counted contract; positive durations/count
Candidate evidence	absolute_source_score; known sources 0; rotation 3 s $\times 2$; pretrigger 0; MAD 3; informed false	counted_candidates; 2; 3 s $\times 3$; pretrigger 6; MAD 3; informed true	counted mode requires sources ≥ 2 and fills = sources -1
Verification/design	convergence hold 2 s; undesired hold 3 s; verify 12 s; design 5 s	same	s; positive
Escape	max 20 s; exit hold 1 s; stall 3 s; min radial progress 0.05 m; approach history 3 s	max 35 s; exit 1 s; stall 3 s; 0.05 m; history 0.5 s	finite positive/nonnegative
Open-field features	assist, approach continuity, interior anchor, active-fill transit, supervisor assist all false; interior min displacement 0.50 m	all five true; 0.50 m	bool; positive m
Recenter	after escape true; bounds true; max 30 s; target clearance 0.05 m; tolerance 0.25 m; hold 1 s; gains 0.50/1.50; caps 0.10 m/s, 0.40 rad/s; rotate threshold $\pi/3$	after escape false; bounds false; other values inert	finite; positive where used
Bounds/directions	room $[-2, 2] \times [-2, 2]$ m; center (0,0); wall margin 0.35 m; lookahead 0.50 m; candidate step $\pi/4$; fill margin 0.10 m; adaptive lookahead false	bounds disabled; fill margin 0.10 m	m/rad/bool

Family	Direct/current default	Selected v8.12 and physical	Constraint/unit
Post-recovery	guidance/recoverable navigation/progress/source handoff/continuity/resume false; retries/refresh 0; window 12 s; min path 0.60 m; max displacement 0.20 m; reversal dot -0.90 ; bypass 0.10 m	disabled	experimental defaults; dot in $[-1,1]$
Freshness/watchdog	pose, sensor, state, command, and recording-ready stale limits 0.50 s; watchdog 20 Hz; zero on shutdown true	same	positive s/Hz; bool
Topics	/odom, source cost, convergence, fill request, supervisor command/stop, state/events/fills, recording ready	same; Vicon separate	resolvable ROS names

9.4 Modified-cost, controller, physical, recorder, and tools

Owner/family	Defaults	Selected value or use	Unit/constraint
Modified cost	all channels true; direction LPF 0.90; min step $1e-4$; affine enabled; gain 0.5; decay $5e-7$; min norm $1e-4$; max age 30 s; sign 1; PDE history true; exclusion factor 3; outside_to_anchor	gain 0.50; decay $5e-7$; max age 35 s; sign 1	finite; LPF $[0,.999]$; s/m as named
Selected controller	config-owned	Directional_Controller; $k_{vx}=1.0$; $k_{wz}=5.0$; wheel radius 0.033 m; separation 0.158 m; max 70 RPM; caps 0.05 m/s and 0.30 rad/s physical	finite hardware limits
Physical time/pose/evaluation	direct wall time; /odom; Vicon disabled	use_sim_time=false; /odom; passive Vicon enabled on evaluation topic	strict role separation
Physical cost	calibration optional/inert	sensor +V, cost -V, source score invalid	volts
record_run	mode required; metadata required; root ~/Experiments/GESC-Gaussian/runs; installed manifest; sqlite3; duration 0; preflight 45 s; ready 15 s; zero 3 s; target exit 15 s; post-zero 0.5 s; ready 10 Hz	physical wrapper supplies metadata/root/target; manual duration	positive time/rate; sqlite3 authority

Owner/family	Defaults	Selected value or use	Unit/constraint
validate_run	one run-directory positional argument	post-run/offline validation	readable finalized run
run_scenario	scenario YAML; required operator; optional repeated case IDs, root, summary, GUI, dry-run	versioned Phase 08 YAML	finite serial scenario contract
analyze_run	run directory; optional output, channel, sync tolerance	one retained run	valid bag/topic/channel
summarize_matrix	one or more inputs; required output directory	compatible versioned summaries only	never pool incompatible gates

Simulation light position/intensity, start pose, bounds, disturbance, and acceptance values are case-owned in versioned scenario YAML rather than global algorithm defaults. The v8.12 visible case used two sources, relative inputs 533.333... and 1600, start (0,0,0), local source (0.883883,0.883883), global source (3.5,3.5), and seed 20001. Relative Gazebo lumen inputs are model parameters, not calibrated physical lux.

10. Implementation chronology: Phases 00–10

The project advanced through evidence-gated phases. The commit column is a navigation boundary, not a claim that every later amendment was present at the first hash. Each linked handoff resolves the exact files, commands, skips, and dirty-state receipt.

Phase	Objective and implemented outcome	Verification/result	Commit boundary and handoff
00	Read-only repository, owner, interface, test, and risk audit. No runtime change.	Baseline collected 885 tests: 875 inherited failures, 1 skip, no collection errors.	baseline e0c693e; handoff
01	Added six typed observability messages and instrumented existing cost/filter/controller/fill owners without changing outputs.	16 focused passed; global 901 with unchanged 875 failures/1 skip.	6eaab86-2ba43f3; handoff
02	Added opt-in eight-state supervisor, switchable component weights, stale/failsafe zero, and compatibility profile selection.	44 focused passed; global 929/875 failures/1 skip.	000f001-8cf93f5; handoff
03	Added synchronized basin estimation, adaptive anisotropic design, residual validation, association/merge revisions, and typed fill lifecycle.	65 focused passed; global 950/875 failures/1 skip.	8717bf5-8789693; handoff

Phase	Objective and implemented outcome	Verification/result	Commit boundary and handoff
04	Added repulsive exit geometry, assisted escape, bounded recenter, supervisor contribution, and command safety. A visible smoke exposed launch DOUBLE typing and SIGINT defects; both were repaired and the failed evidence retained.	93 focused and visible startup/SIGINT passed; global 978/875 failures/1 skip.	b0e024d–2e40388; handoff
05	Unified metadata-rich sqlite3 recording, topic manifest, readiness gate, final-zero ordering, and sole validator.	113 focused passed/1 skipped; a 19-topic bag, final zero, provenance, and validation passed; global 999/875 failures/2 skips.	3982eb9–a7df8cf; handoff
05.5	Consolidated Phase 00–05 knowledge and hardened Plan/status/check-point/handoff recovery. No runtime change.	Phase 05 accepted run and totals revalidated.	7cc9ace; handoff
06	Added deterministic YAML scenario schema/runner, seeded Gazebo execution, cleanup, and acceptance evaluation.	143 focused/2 skips plus one recorded E2E pass; robust and legacy recording/cleanup 2/2; two short-run ground-truth misses were nonpredicates; global 1027/872 failures/3 skips.	b84518a–8c6191d; handoff
07	Added bag-authoritative synchronization, metric tables, plots, failure analysis, CSV compatibility export, and matrix summaries.	153 focused/2 skips plus 10 new tests; one complete and one partial bag; 22 CSV and 16 PNG products; global 1037/872 failures/3 skips.	80d5c6a; handoff
07.5	Qualified deterministic contacts, noise, sensor delay, and pose delay as Phase 08 prerequisites.	162 focused/2 skips; five probes complete; global 1031/856 inherited failures/3 skips. No robustness claim.	2a74001; handoff
08	Executed versioned simulation development, corrections, selected reproductions, and frozen gates. Many genuine and infrastructure failures were retained.	Broad readiness failed. Scoped maxima were v8.10 primary 11/11, v8.11 secondary 6/6, and v8.12 visible 14/14; v8.12 varied case 13/14 and three withheld.	terminal receipts through c04c222; 08.8 handoff

Phase	Objective and implemented outcome	Verification/result	Commit boundary and handoff
09	Backed up and reconciled a physical source snapshot; ported shared owners/interfaces; added hardware/evaluation/readiness/rotation adapters; repaired seven observed commissioning faults; retained eight numbered trials.	Host/static/check-only gates passed at their scopes. Eighth run was first physical two-basin behavioral success but stopped before second confirmation/goal and passed 61/62 completeness. No broad physical-ready result.	terminal V1 HEAD e3dd0ef; handoff
10	Audited every package/phase/result artifact; wrote this master report, coverage proof, current navigation, validation record, checkpoint, and handoff. Documentation only.	Acceptance is the Phase 10 documentation gate, not a runtime/robustness rerun.	parent e3dd0ef; closeout is the commit containing this report with the authorized subject above.

There were no Phase 00–07 live status/checkpoint files because that durable policy was introduced later; the absence is a historical workflow gap, not an unrecorded runtime implementation. Their Plans, handoffs, bridge, tests, and Git receipts remain the controlling evidence.

11. Simulation scenarios, methods, and complete V1 results

11.1 Method and denominator rules

run_scenario validates schema, resolves a frozen profile, creates a unique run ID, invokes the sole recorder around the canonical Gazebo graph, enforces finite wall/run/shutdown budgets, validates the finalized run, evaluates only the case's declared predicates, and writes a scenario summary. Seeds, source geometry/intensity, start pose, disturbances, algorithm overrides, and exact success predicates live in the versioned YAML. GUI probes were visible; suites were bounded and generally headless. Failed, contaminated, infrastructure- invalid, early-stopped, and withheld cases remain in their declared denominators.

Development probes answer a local diagnostic question. Selected reproduction sets measure a known-success population. Prospective fixed-layout repeats measure only that layout/version. Formal pass means all predicates in that version; scientific/behavioral pass can coexist with an evaluator/evidence false negative but is never relabeled formal. Results from versions that changed controller, evaluator, topology, or evidence contracts are not pooled.

11.2 Phase 08 v1–v6

Version/probe	Executed result	Failure/correction boundary	Primary evidence
v1 training	81/81 ran; C0–C5/C7/C8 infrastructure-eligible, C6 ineligible; every candidate had 0 end-to-end successes and 0 escapes. C8 was chosen only by lowest median path 13.2286087344 m.	Selection was not behavioral success.	parameter selection

Version/probe	Executed result	Failure/correction boundary	Primary evidence
v1 holdout	12/12 ran: controller goal 0/12, ground truth 5/12, failsafe 2/12, collision 0, timeout 0.	Activation timing unreachable; goal verification was orientation dependent and instantaneous.	failure closeout
v1 pass 1	Planned 519; stopped at 203 retained directories: 200 recording-complete, 3 failed; current root has 320 bags.	No pass_1.json; passes 2–3 never began. Immutable failed version.	failure closeout
v2 activation	10/10 ran; complete lifecycle 1/10, recording/analysis 3/10, cleanup 10/10, goal 1/10, valid no-collision 9/10, seven typed fills, one escape.	Six stale-source fill timestamps plus missing publisher snapshot/lifecycle evidence. Gate 2 failed; 30 tuning, 20 holdout, 50 validation, 10 repeats withheld.	validation , failure
08.1 high-goal	Reached SEARCH, then VERIFY_EXTREMUM, then GOAL_HOLD.	Diagnostic activation only.	activation contract
08.1 fill	Fill about 0.09 s, repulsive escape 8.714 s, then recenter orbit and 30 s failsafe.	Activation subclaim passed; full diagnostic failed.	activation contract
08.2 recenter	Seed 8304: one fill, escape 7.532641327 s, recenter 8.610255022 s, returned SEARCH, no collision/failsafe/evidence defect.	Sealed recenter-only success; not v3 acceptance/readiness.	handoff
v3A	One GUI seed 9301; recording/cleanup passed.	105 non-ground contacts came from an accidentally enabled positive-control obstacle at the robot pose; contaminated; nine withheld.	failure
v3B	Seed 9301 infrastructure/safety and controller/ground-truth goals passed; final distance 0.039083 m.	Full fill/escape/recenter occurred where direct/no-recovery was required: genuine behavioral miss; nine withheld.	failure
v3C	Required seed 9302 failed before Gazebo.	Incorrect executor/context caused <code>AttributeError: __enter__</code> ; no behavioral run.	prelaunch failure
v3D	v3B carried plus one new GUI 9302; robot moved through second fill/failsafe; bag had 400,131 messages.	Recorder SIGINT invalidated context before readiness-false/final-zero/finalization; infrastructure-invalid; eight withheld.	failure

Version/probe	Executed result	Failure/correction boundary	Primary evidence
v4	Three Gazebo executions; corrected r2b seed 10601 was sole eligible 1/120, directly reached goal at 0.096979 m with recording, cleanup, final zero, and no collision.	Required fill/escape/recenter never occurred. Earlier run lost finalization; r2 packaging ran no Gazebo. Later stages withheld.	validation , erratum
v5/v5b	No valid behavioral Gazebo result.	Recovery-lineage prelaunch and stale-shell recorder defects; superseded by v5c.	v5 report
v5c	Eight visible seeds; recording/cleanup 8/8; four completed a lifecycle/goal path.	Declared blocker encounter 0/8; fill-to-blocker 1.2624--1.5782 m, required ≤ 0.35 m. Wrong basin; 112/120 withheld.	validation
v6 sweep	H25 primary/full pass; H50 stall/reject/failsafe; H70 local treated as goal/no fill; H85 primary pass then reject/failsafe.	H25 ratio selected for repeats only.	validation
v6 repeats	Three H25: one full pass; two later rejection/failsafe.	Full repeatability 1/3, required 2/3; first recovery 2/3.	validation

11.3 Phase 08.7 correction probes and reproduction

Milestone	Exact result	Status/limitation	Evidence
M1	Historical immutability/no-Gazebo qualification passed.	Infrastructure only.	JSON
M2	Visible 18001 Stage A/fill passed; Stage B best 2.204118 m, required ≤ 0.35 m.	Failed; later reject/failsafe.	report
M2.1	Infrastructure passed; SEARCH for 360 s, no convergence/fill.	Failed; efficiency cap 0.35 overconstrained. A prior shell <code>set -u</code> error was not an attempt.	report
M2.2	Detector/fill plus direct, assisted, recenter, SEARCH recovery occurred.	Formal Stage A false due direct-only reporter; monitor never armed; 0.60 m geometry too strict before failsafe/contact.	report
M2.3	18001 passed development contract; Stage B 1.199192 m under 1.20 m.	One development probe only.	report
M3	Five positions, infrastructure 5/5, combined 1/5.	One pass; one 39/40 sample miss; wall collision; wrong local; recenter timeout. Required 5/5 failed.	report

Milestone	Exact result	Status/limitation	Evidence
M4	18201 behavior passed, Stage B 1.199229 m.	Formal failure solely from merged /joint_states publishers' 225 ms timestamp regression; suite withheld.	report
M4.1	Timestamp correction verified.	18207 Stage B failed at 2.399676 m after post-recovery looping; suite withheld.	report
M4.2	18208 behavior passed, Stage B 1.198171 m.	Completeness 47/48: two supervisor configuration events lacked attribution; suite withheld.	report
M4.3 visible M4.3 suite	18308 full pass. Eight ran; infrastructure 8/8, Stage A 7/8, formal 6/8.	Authorized suite only. r1.0 unsafe direction; r1.5/22.5 wrong departure. Optional three-light withheld.	visible report suite
M4.4 visible M4.4 suite	18408 assisted path passed, Stage B 1.198640 m. Eight: formal 5/8; behavior 5/7 eligible; one infrastructure-invalid.	Authorized corrected suite only. Controller load failure, exact-reversal miss, and fixed-horizon budget defect.	suite/report suite
M4.5	18508 Stage A/one fill passed.	Stage B 3.628273 m; reversal threshold -0.90 missed observed -0.841 .	report
M4.6	Threshold -0.80 armed.	Tangent made zero source progress; premature clearance release; Stage B 2.617347 m.	report
M4.7	Stage A passed.	Stage B 2.744859 m; direction truly opposed motion so new corridor correctly did not arm.	report
M4.8	17 evidence-selected historical successes rerun: formal 13/17, behavioral 14/17, Stage A+fill 16/17, Stage B 14/17, recording 16/17; safety/final-zero/cleanup/sqlite 17/17.	Retrospective selected set; three behavioral disagreements, one console-clean miss; all 16 recoveries direct, no assist.	report , manifest

11.4 Phase 08.8 v8.0–v8.12

Version	Executed result	Formal/scientific boundary	Evidence
v8.0	Visible 18801 complete, first unknown candidate and one fill.	0/1: failed departure, returned/revisited; later gates withheld.	report

Version	Executed result	Formal/scientific boundary	Evidence
v8.1	18901 assisted recovery and global arrival, 0.155548 m.	0/1: correct second candidate rejected by post-confirmation-only evidence slice.	report
v8.2	Visible 19001 formal pass; repeat 19011 ran.	Formal 1/2: 19011 Stage B 3.180001 m from weak/inconsistent fill; seeds 19012–19020 withheld.	probe , repeats
v8.3	Visible 19101 and repeats 19111–19114 passed.	5/6: 19115 wrong escape alignment and Stage B 5.167727 m; remaining withheld.	repeats
v8.4	Visible 19201 and 19211 passed.	2/3: 19212 revalidator reversed selected direction; Stage B 4.679383 m; remaining withheld.	repeats
v8.5	Visible 19301 and repeats 19311–19315 passed.	6/7: 19316 command interaction reversed actual exit; Stage B 2.149315 m; remaining withheld.	repeats
v8.6	Visible correction passed; 19411 scientifically reached global at 0.123614 m.	Formal 1/2, scientific 2/2; 19411 was 12/14 due one cross-topic transition sample and external echo cleanup contamination; remaining withheld.	repeats
v8.7	Visible and 19511–19513 passed; 19514 reached global at 0.112068 m.	Formal 4/5; 19514 was 13/14 from evaluator assist-entry false negative; remaining withheld.	repeats
v8.8	Visible and 19611–19615 passed; 19616 scientifically passed direct, final 0.104056 m.	Formal 6/7; fixed assist-only evaluator rejected valid direct route; remaining withheld.	repeats
v8.9	19701 dispatched; no Gazebo, bag, or plots.	Pre-Gazebo failure: installed runner's /tmp cwd made recorder Git lookup fail. No repeats.	report
v8.10 primary	Visible 19801 plus 19811–19820 all passed.	11/11 formal , fixed primary layout only.	probe , repeats
v8.10 secondary	19851 scientifically complete direct route, final 0.105549 m.	Formal 10/14 and 0/1 because evaluator used declared lamp at 0.60 m rather than aggregate basin at 0.75 m. Repeats withheld.	report
v8.11 secondary	Evaluator-only topology correction; visible 19901 and repeats 19911–19915 all passed.	6/6 formal , five direct and one assisted; exact layout only.	visible , repeats

Version	Executed result	Formal/scientific boundary	Evidence
v8.11 matrix	Only seed 19931 ran.	0/1 formal/science: immediate failsafe because pre-fill odometry did not exceed exit radius; three seeds withheld.	report
v8.12 visible	New default-off interior_farthest; seed 20001, assisted, final 0.1167065 m.	14/14 formal ; one selected visible case only.	report
v8.12 matrix	Only 20031 ran; direct scientific path, one fill, final 0.127721 m.	13/14 formal , 1/1 science: exit alignment 0.735563 < 0.80. Seeds 20032–20034 withheld; no v8.13.	report
Optional three-light	0 executions.	Not run; prerequisites never authorized it.	final report

11.5 Retained simulation products

The 56 current Phase 08 run roots under /home/mattb/Experiments/GESC-Gaussian/runs contain 459 sqlite3 bag files, 456 scenario_result.yaml files, and 1,903 PNGs. These are current local counts, not an acceptance denominator. The terminal selected roots include:

- /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_10_primary_repeats;
- /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_11_secondary_repeats;
- /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_12_interior_anchor_probe;
- /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_12_broad_matrix; and
- /home/mattb/Experiments/GESC-Gaussian/runs/phase08_v7_success_reproduction_1.

Selected terminal summaries and plot roots verified locally during the Phase 10 audit are:

- v8.10 primary visible: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_10_primary_probe/phase08_v8_10_primary_visible_probe_summary.yaml;
- v8.10 primary repeats: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_10_primary_repeats/phase08_v8_10_primary_repeats_summary.yaml;
- v8.10 secondary: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_10_secondary_probe/phase08_v8_10_secondary_visible_probe_summary.yaml;
- v8.11 secondary visible: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_11_secondary_probe/scenario_summaries/20260731T210056130307Z_phase08_v8_11_secondary_visible_probe.yaml;
- v8.11 secondary repeats: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_11_secondary_repeats/scenario_summaries/20260731T211244723799Z_phase08_v8_11_secondary_repeats.yaml;
- v8.11 matrix: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_11_broad_matrix/scenario_summaries/20260731T215507749994Z_phase08_v8_11_broad_matrix.yaml;
- v8.12 visible summary: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_12_interior_anchor_probe/scenario_summaries/20260731T231350696332Z_phase08_v8_12_interior_anchor_visible_probe.yaml;
- v8.12 visible plots: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_12_interior_anchor_probe/2026-07-31/20260731T231351648763Z_simulation_phase08_v8_12_interior_anchor_visible_probe-v8_12_interior_anchor_r1p25_a45_rati_439a8eac/analysis/phase07/plots;
- v8.12 matrix summary: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_12_broad_matrix/scenario_summaries/20260731T233137289914Z_phase08_v8_12_broad_matrix.yaml; and
- v8.12 matrix plots: /home/mattb/Experiments/GESC-Gaussian/runs/phase08_8_12_broad_matrix/2026-07-31/20260731T233137972843Z_simulation_phase08_v8_12_broad_matrix-v8_12_matrix_r1p25_a45_ratio1to3_20031-robust_gaussian_6efe976a/analysis/phase07/plots.

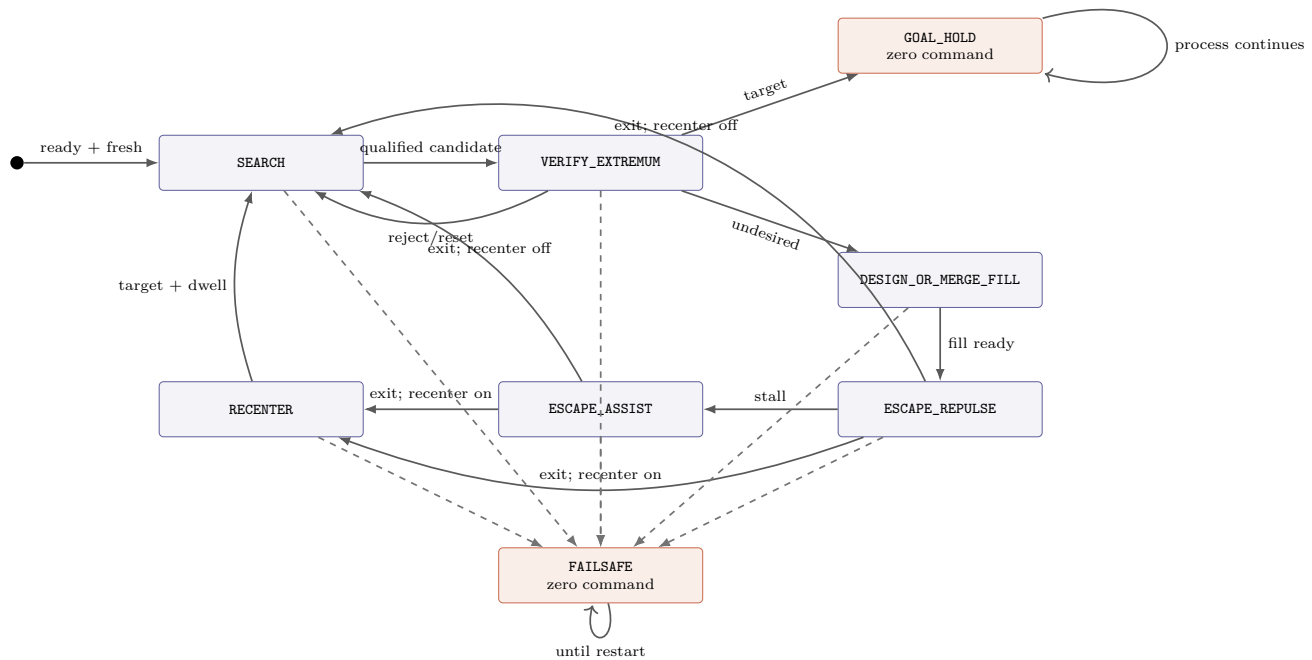


Figure 3: Eight-state robust-supervisor graph. Dashed transitions represent stale/invalid-input or timeout paths to FAILSAFE; the following table retains the complete per-state transition contract.

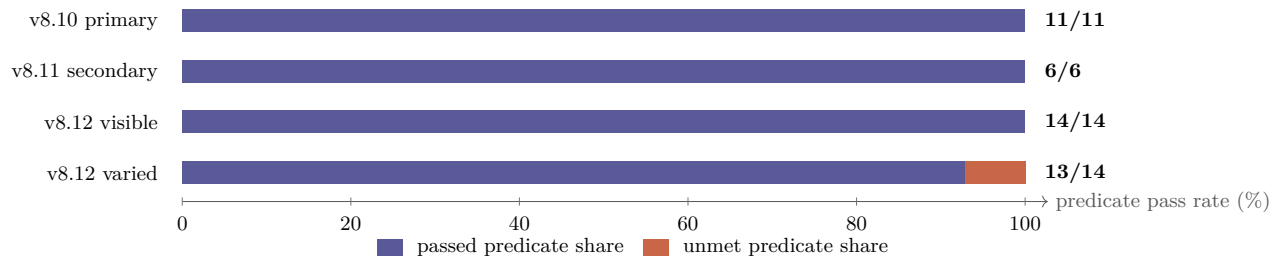


Figure 4: Scoped terminal simulation predicate outcomes. Each bar is normalized within its own named contract; the raw ratios are printed at right. These selected cases have different denominators and are not pooled into a broad robustness rate.

Each analyzed Phase 08.8 run normally retains nine standard plots: trajectory_sources_fills.png, candidate_ranking.png, cost.png, components.png, state_events.png, weights.png, command_saturation.png, radial_escape.png, and gaussian_history.png. Availability of a plot is evidence-product completeness, not behavioral pass.

The terminal outcome is unambiguous: no version completed the required 70-unique-run denominator, no simulation-ready tag exists, and the selected version/layout denominators above must remain separate.

12. Physical integration, commissioning, and complete V1 results

12.1 Snapshot, parity, and operator architecture

Phase 09 first created a recoverable snapshot backup and SHA-256 inventory, classified the intended shared, adapted, and physical-only files, ported the six typed interfaces and shared algorithm owners, added physical adapters without forking the algorithm, and qualified legacy assets. Transfers were bounded, manifested, and rollback-backed. Source/Pi parity was a prerequisite, never a claim of hardware readiness.

The current local snapshot contains 347 regular files, zero symlinks, and zero cache directories. The final M8L inventory convention has 435 rows: those 347 files plus 88 directory rows including the root. Relative to M0, 41 files were added, 19 modified, zero deleted, and 287 unchanged. The final retained M8L receipt is under /home/mattb/physical_TB3_files_snapshot/phase09_backups/20260805T002822-0700_m8l_rotation_initialization_rearm/. The tracked 345-file/432-row M8B manifests are historical, not the final M8L inventory.

The selected no-lidar TurtleBot3 base owns wheel/IMU /odom; the photoresistor adapter emits +V sensing and -V minimization cost; shared GESG/Gaussian owners consume /odom; controller_node alone owns /cmd_vel; Vicon is passive evidence; record_run alone owns readiness, rotation authorization, recording, validation sequencing, and final zero; and phase09_rotation_node accepts only that recorder authorization.

12.2 M0–M8L integration chronology

Per-command test counts overlap and are not summed.

Milestone	Result and selected verification	Evidence
M0	Baseline 306 files/389 inventory rows/0 symlinks; 390-entry archive; backup/hash/inventory passed.	backup receipt
M1	52 source rows classified; 51 unique transfer paths; identical parser excluded.	shared manifest
M2	Six interfaces plus 18 shared runtime files; 27 paths byte-identical. Two-package build, shared 300, observability 13 (2 deselected), legacy 34 passed.	static qualification
M3	Adapter/compatibility layer. Accepted checks: Bash 27, launch 6, assets 40, installed focused 69, three-package build 12.6 s.	legacy compatibility
M4	Physical recorder/readiness/final-zero integrated. Physical 36; combined 84 passed/1 skipped; build 13.4 s.	handoff
M5	Selected schema/profile and inert preflight: focused 76; Bash 27; primary/secondary inert preflights exited 2 as expected.	status

Milestone	Result and selected verification	Evidence
M6	Broad host gate: focused 76, shared 272, observability 13/2 deselected, recording 70/1 skipped, legacy 34, Bash 27, show-args 6, interfaces 6, build 13.3 s. Non-gating style still had 917 failures including 873 Q000.	status
M7/M7.1	Transfer seal 339/425, 51-path recovery pass; reseal Phase09 93, shared 272, recording 70/1 skip, observability 13/2 deselected, legacy 34, show-args 6, legacy hashes, build 13.1 s.	Pi transfer receipt
M8A	Direct snapshot qualification: 93 direct, parity/launch 44, Bash 27; 339/425 parity. No on-Pi build/check-only.	handoff
M8B	Expanded snapshot 345/432; focused 309, shared 272, recording 70/1 skip, observability 13/2 deselected, legacy 34, parsing 11 XML/7 YAML/68 JSON, critical 111, hash groups 40/40 and 68/68, build 13.0 s.	status
M8C	Simplified operator contract 342/430; focused 196; ros_esc 153/3 deselected; vehicle 71/3 deselected; Bash 27; Python 9; structured parses; build 12.6 s.	handoff amendment
M8D	Legacy CSV export 343/431; source and installed 209; package subsets; build 12.6 s.	status
M8E	Real-Pi underlay/build/no-lidar repair 345/433; host 22, installed/device 233, vehicle 74, helper lint pass, host build 15.1 s; operator check-only three packages passed in 1m39s.	runtime repair
M8F	CLI/runtime graph repair: focus 10, legacy 37, Vicon/launch 35, full physical 238, build 12.5 s; check-only 14.4 s pass.	first repair
M8G	Startup grace repair: focus 23, Phase09 239, legacy 37, build 15.2 s; check-only 16.3 s pass.	second repair
M8H	Wall-clock initialization repair: clock 16, shared 114, canonical build 14.3 s, installed probe pass, snapshot 256/build 14.9 s, mounted AST/lint 8/8; check-only 16.4 s pass.	third repair
M8I	Recorder grace/rotation evidence: targeted 11, focused 160, related 225, snapshot 264, recorder 69, legacy/recording 38/1 skip, build 14.9 s; check-only pass.	fourth repair

Milestone	Result and selected verification	Evidence
M8J	Recorder gate callback lane separated: focus 2, starvation 5/5, snapshot 266, recorder 69, legacy/recording 38/1 skip, build 14.8 s; check-only pass.	fifth repair
M8K	Heartbeat/safety/passive lanes separated: focus 4, starvation 5/5, recorder-file 143, snapshot 268, recorder 69, legacy/recording 38/1 skip, build 14.9 s, installed 4; check-only pass.	sixth repair
M8L	Rotation initialization re-arm: focused 16, changed-file 165, snapshot 269, recorder 69, legacy/recording 38/1 skip, build 15.4 s, installed 4; pre-run check-only passed.	seventh repair

12.3 Retained commissioning attempts

An unnumbered M8E commissioning attempt failed before the eight attempts below from stale underlay/install/serial-flag handling. Its recorded Pi path is /home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-04/20260804T152851316144Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_df84804a.

Attempt	Exact retained outcome	Evidence
1	Controller/filter rejected ROS arguments; 3,187 messages; readiness never true; no command/source/filter/timekeeper, no motion; 18 completeness failures.	first repair
2	Startup-grace mismatch; 10,234 messages; readiness false; 1,202 /cmd_vel, all zero; no source cost/timekeeper/motion; target and bag clean.	second repair
3	Late use_sim_time change caused InvalidHandle; 8,833 messages; Vicon/odom live; 815 commands all zero; 421 rotation authorizations false; no RPM/motion.	third repair
4	First readiness and bounded physical SEARCH, about 10.6 s and 0.13 m; no fill. Recorder falsely aged source/filter to 0.577/0.573 s while bag gaps were <0.25 s; zero/cleanup passed.	fourth repair

Attempt	Exact retained outcome	Evidence
5	Readiness 6.189497 s; net 0.045709 m odom/0.059153 m Vicon; no fill. Gate callbacks paused ~1.572 s; 0.5 s rotation lease faulted/zeroed; base zero 12.663 ms; export passed, completeness failed.	fifth repair
6	Readiness 119.608472 s; odom path/net 4.554253/2.461536 m; Vicon 5.282798/2.329767 m; convergence after 6.003336 s at $(-1.492954, -1.607303)$. Operator stopped 5.4 s into 9 s verification; no fill; 48,731 messages; zero/export passed, completeness failed.	sixth repair
7	Barrier/auth passed, but first authorization initialized hardware and was consumed before re-arm; stale fault ~0.5675 s later; readiness false; 1,900 commands all zero; no source/filter/timekeeper/motion/fill.	seventh repair
8	First narrow physical two-basin behavioral success: convergence, 9 s classification, one fill $(2.227914, 0.264837)$, amplitude 0.562125, exit radius 1.366771 m; repulse+assist $\rightarrow$ SEARCH; no recenter; odom path/net 7.797691/4.197350 m; Vicon 9.027076/4.088632 m. Manual stop before second convergence/goal.	eighth validation

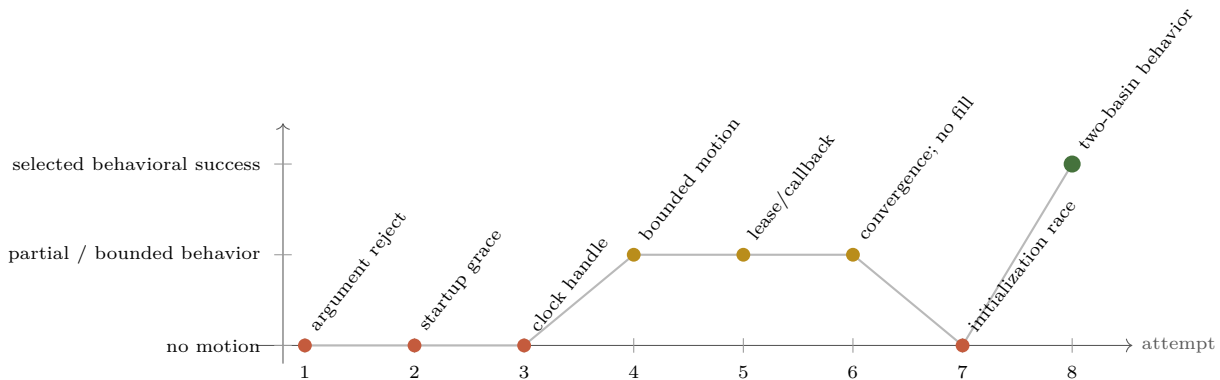


Figure 5: Phase 09 numbered-attempt progression. Vertical levels are explicit evidence categories, not a continuous performance score. Attempt 8 is a narrow behavioral success and remains incomplete for second convergence, GOAL_HOLD, and broad physical readiness.

The exact numbered run roots are retained in the Phase 09 repair reports and coverage index. They are Pi-side paths

quoted from repository evidence; Phase 10 did not access the live mount to revalidate their presence.

For lossless recovery, the eight quoted run roots are:

Attempt	Retained Pi-side path
1	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-04/20260804T201250796597Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_6d9b4ac3
2	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-04/20260804T205606472031Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_d4f0178d
3	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-04/20260804T212226731572Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_309b9e71
4	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-04/20260804T221143126652Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_b27ffe0
5	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-04/20260804T224523162901Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_8d7f9b77
6	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-04/20260804T231459927630Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_2155c436
7	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-05/20260805T000043179553Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_d9eb86a8
8	/home/pi/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/2026-08-05/20260805T004605139747Z_physical_phase09_selected_primary_r1p5_a45_ratio1to4_cd83f78f

12.4 Eighth-run claim and 61/62 limitation

The eighth bag was 186,003,456 bytes. It showed one local candidate, a fill, repulsive then assisted escape, return to SEARCH, and stronger-light reacquisition. Five later rotation-scale voltage peaks had median 1.9013 V, 4.37 times the local candidate estimate near 0.435 V. These observations support progress from a weaker basin toward a stronger signal.

They do not support complete second-extremum acceptance: there was no second convergence confirmation, second classification, or GOAL_HOLD; the operator ended the open-field run manually. Completeness was 61/62. After the first false rotation gate, one already-in-flight 20 RPM RUNNING status arrived for about 45 ms. Base zero appeared in about 6 ms, rotation zero in about 67 ms, and revoked FAULT/zero in about 92 ms. The current receipt-ordered validator calls this a failure even though the bounded shutdown behavior was observed. M8M is documented as a possible future validator correction but was not implemented. V1 therefore closes as selected behavioral success with an evidence-order false negative, not formal physical acceptance. The [terminal Phase 09 checkpoint](#) records that exact M8L/eighth-run boundary.

13. Recording, analysis, and artifact products

`ros2 run ros_esc record_run` is the sole recorder for simulation and physical runs. It creates a unique date/run directory, copies and resolves metadata, captures branch/commit/dirty provenance, snapshots parameters and the graph, starts `ros2 bag record` with the installed topic/QoS manifest, gates motion until readiness, retains target/recorder consoles, and finalizes integrity and completeness records. `ros2 run ros_esc validate_run RUN_DIRECTORY` is the sole validator. A second recorder, live CSV authority, or console-only substitute would violate the evidence contract.

The shutdown transaction is intentionally ordered:

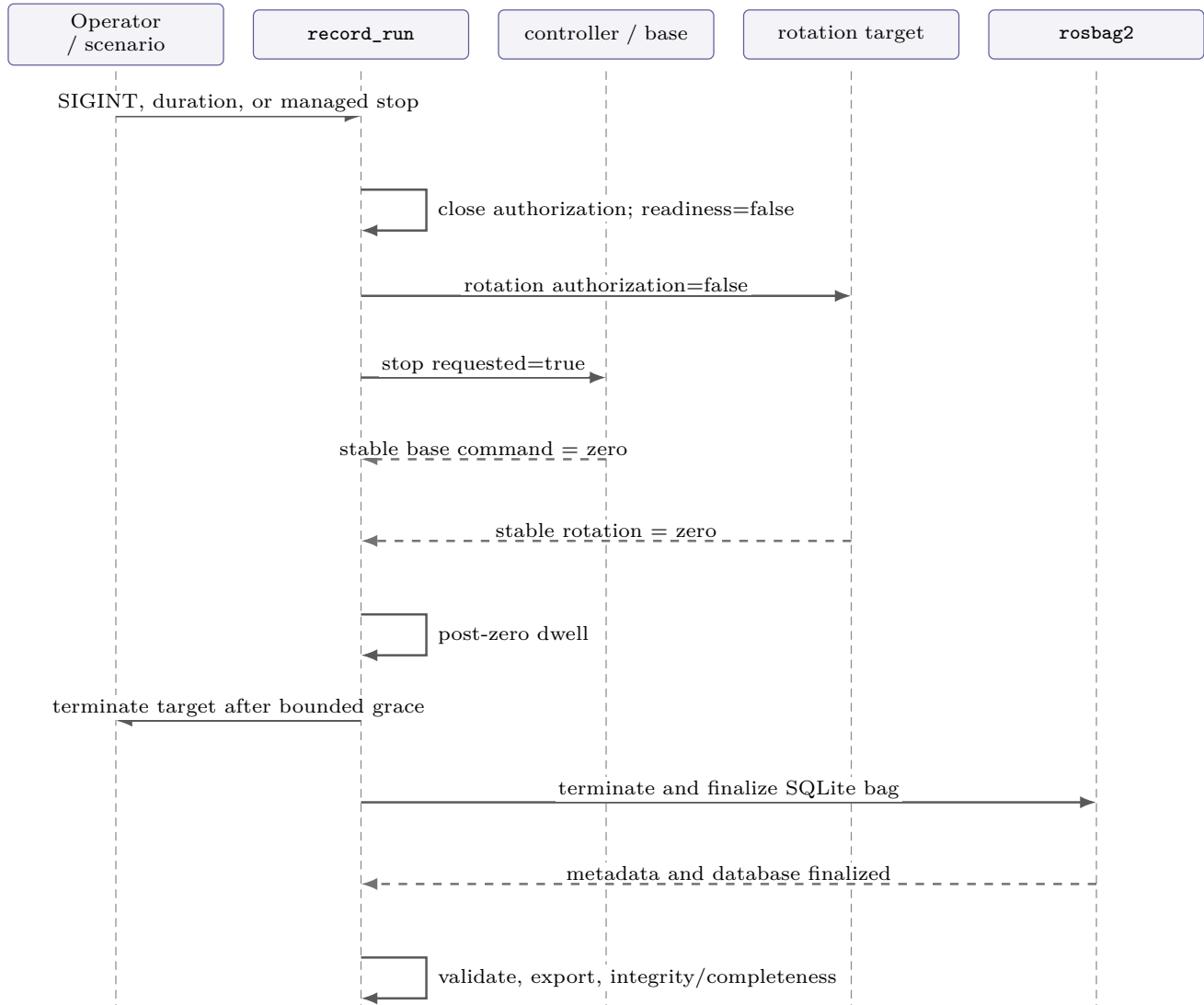


Figure 6: Managed recorder shutdown transaction. Authorization closes before target termination; stable base and rotation zero are observed before the bag is finalized and derived products are written.

Typical retained products are:

Product	Authority and purpose
metadata.yaml and resolved configuration/parameter snapshots	Operator intent plus exact execution provenance. A desired value is not accepted when the live resolved value disagrees.
sqlite3 bag (*.db3 plus metadata)	Primary time-series authority. Required typed and compatibility topics, shutdown evidence, and evaluator streams are read here.

Product	Authority and purpose
completeness.json / integrity result	Machine-checkable required topics/types/counts/rates, final zero, readiness, stop, configuration, and lifecycle checks.
target/recorder console	Diagnostic context only; cannot replace required bag data.
scenario_definition.yaml, resolved_scenario.yaml, scenario_result.yaml	Original, resolved, and observed simulation case contract/outcome.
synchronized analysis tables and summaries	Derived, traceable products with validity/provenance columns; never override raw bag evidence.
nine standard PNG plots	Trajectory/fills, candidate ranking, cost, components, state/events, weights, saturation, radial escape, Gaussian history.
matrix summary	Aggregates only compatible run/version contracts and preserves unavailable/failed rows.
physical familiar CSV export	Post-finalization, atomic and idempotent compatibility views; the bag remains authority.

The physical export creates encoder.csv, cost_value.csv, filter_value.csv, control_value.csv, and evaluation-meaning odometry.csv, plus raw_cost_value.csv (-V) and algorithm_odometry.csv (/odom). legacy_csv_manifest.json records aliases, semantics, row counts, sizes, hashes, and export errors. It does not fabricate comments.txt, and old browsers may not auto-discover the nested run root.

Simulation runs normally live under ~/Experiments/GESC-Gaussian/runs/<date>/<run-id>/; scenario Plans may select a version-specific root below .../runs. The selected physical wrapper uses \${HOME}/turtlebot_rotating_sensor_tests/gesc_gaussian_two_source/<UTC-date>/<run-id>/ on the Pi. Raw bags and high-volume plots remain outside Git. Git retains Plans, schemas, summaries, manifests, validation reports, hashes, and exact paths. Failed and partial run directories are preserved; cleanup or storage work requires a separate read-only inventory and explicit deletion authority.

14. Testing and validation evidence

Test counts below are per recorded gate. Global counts contain an inherited style baseline and are not a measure of behavioral quality. Focused suites grew as capabilities were added; later counts often include earlier tests and must not be summed.

Boundary	Focused/runtime evidence	Global or known debt	Claim
Phase 00	Audit only.	885 total; 875 failures; 1 skip; failures dominated pre-existing style/lint.	Baseline, not implementation acceptance.
Phase 01	16 passed.	901 total; 875 failures; 1 skip.	Typed observability compatibility.
Phase 02	44 passed.	929 total; 875 failures; 1 skip.	State/profile/safety core.
Phase 03	65 passed.	950 total; 875 failures; 1 skip.	Pure estimator/designer/registry plus ROS wiring.
Phase 04	93 passed plus visible Gazebo startup/SIGINT smoke.	978 total; 875 failures; 1 skip.	Escape/recenter and bounded lifecycle correction.
Phase 05/05.5	113 passed, 1 skipped; complete 19-topic recorded run.	999 total; 875 failures; 2 skips.	Recorder/validator/final-zero.

Boundary	Focused/runtime evidence	Global or known debt	Claim
Phase 06	143 passed, 2 skipped; bounded recorded E2E passed.	1,027 total; 872 failures; 3 skips.	Scenario infrastructure, not robustness.
Phase 07	153 passed, 2 skipped plus 10 new; complete and partial bags analyzed.	1,037 total; 872 failures; 3 skips.	Analysis/plot/CSV semantics.
Phase 07.5	162 passed, 2 skipped; five prerequisite probes.	1,031 total; 856 inherited failures; 3 skips.	Contact/noise/delay support only.
Phase 08	Many version-specific static, runner, recorder, analyzer, GUI, and headless gates; exact results in Section 11.	No 70-unique acceptance denominator; broad gate failed.	Selected/versioned evidence only.
Phase 09 M6	Focused 76; shared 272; observability 13/2 deselected; recording 70/1 skipped; legacy 34; Bash 27; launch args 6; interfaces 6; build 13.3 s.	Non-gating style 917 failures including 873 Q000.	Host qualification, not hardware behavior.
Phase 09 M8L	Focused 16; changed-file 165; snapshot 269; recorder 69; legacy/recording 38/1 skipped; build 15.4 s; installed 4; check-only passed.	Eighth run 61/62 completeness.	Selected physical behavior, not formal/broad acceptance.
Phase 10	Context, required-doc, TSV coverage, headings, links, fences, structured file parsing, interfaces/entry points, claim searches, path scope, and Git whitespace checks.	No ROS build, Gazebo run, matrix, Pi access, or hardware run: documentation-only by Plan.	Exact commands/outcomes are in the Phase 10 validation record .

Infrastructure completeness and behavior remain distinct. A clean bag can record a failed algorithm; a successful trajectory can be formally invalid when required evidence is missing; and a static/check-only pass cannot prove motion. Skips and deselections were retained where environment, style debt, or test design made them intentional. Phase 10 did not rerun historical matrices to manufacture a cleaner denominator.

15. Verified operating instructions

These commands are documentation. Phase 10 did not execute Gazebo or physical hardware. Operators must first verify their environment, branch, metadata, workspace cleanliness, and physical authorization.

15.1 Build and source

```
cd /home/mattb/dsim-lab/ros2_ws
source /opt/ros/humble/setup.bash
colcon build --symlink-install \
  --packages-select ros_esc_interfaces ros_esc turtlebot3_rotating_sensor
source install/setup.bash
```


15.2 Inspect or run a bounded simulation case

Dry-run validates and expands without ROS/Gazebo:

```
ros2 run ros_esc run_scenario \
  src/ros_esc/ros_esc/scenario_runner/scenarios/phase06_smoke.yaml \
  --operator "$USER" \
  --dry-run
```

For a visible, finite one-off execution, add --gui and use only a currently approved scenario/case/root. Do not resume closed Phase 08 roots or dispatch a withheld matrix. A direct robust launch, useful for inspection rather than acceptance, is:

```
ros2 launch turtlebot3_rotating_sensor gazebo.launch.xml \
  gazebo_gui:=True \
  algorithm_profile:=robust_gaussian_v1 \
  use_pde_extensions:=True \
  enable_observability:=True \
  recording_ready_required:=False
```

Direct launch is not a managed evidence run. For one, wrap the target exactly as an argument vector after --:

```
ros2 run ros_esc record_run \
  --mode simulation \
  --metadata-input src/ros_esc/test/fixtures/recording_smoke_metadata.yaml \
  --duration-sec 8.0 \
  --preflight-timeout-sec 60.0 \
  --runs-root ~/Experiments/GESC-Gaussian/runs \
  -- \
  ros2 launch turtlebot3_rotating_sensor gazebo.launch.xml \
  algorithm_profile:=robust_gaussian_v1 \
  use_pde_extensions:=True \
  enable_observability:=True \
  recording_ready_required:=True
```

The fixture is for a smoke test; a research run requires a copied, honestly completed metadata template and a versioned scenario/Plan.

15.3 Validate and analyze retained evidence

```
ros2 run ros_esc validate_run /absolute/path/to/run

ros2 run ros_esc analyze_run \
  /absolute/path/to/run \
  --output-dir /absolute/path/to/run/analysis

ros2 run ros_esc summarize_matrix \
  /absolute/path/to/compatible/summary-or-run ... \
  --output-dir /absolute/path/to/matrix-summary
```

Do not combine incompatible experiment versions. An unavailable external path must be labeled unavailable, not reconstructed from prose.

15.4 Selected physical operator procedure

The current selected procedure is the dedicated wrapper in the physical Pi workspace. It builds/sources the required packages, checks installed/source parity and distinct devices, launches no-lidar base and the selected graph, creates a run root, starts the sole recorder, and prints bounded diagnostics. The operator command recorded by the current checklist is:

```
cd ~/ros2_ws/src/turtlebot3_vehicle_nodes/bash_scripts/light_esc_experiments/voltage_cost_values
./gesc_gaussian_two_source_voltage.bash
```

Before using it, follow the lab power, e-stop, open-field-clearance, device, OpenCR, photoresistor, Vicon-evaluation, and clock SOP in [the physical-readiness document](#). Do not change the Pi clock after ROS/rosbag starts. Vicon may be absent/stale only as an explicitly failed evaluation-completeness condition; it cannot be substituted for /odom or used to gate motion.

Stop normally with Ctrl+C. Do not treat GOAL_HOLD as process exit. Leave the terminal running while readiness/rotation are revoked, final zero is observed, the target and bag finalize, validation runs, and CSV export completes. On any fault, retain the run and diagnose it before retrying. Phase 10 did not run this procedure, access the Pi, or authorize another trial.

16. Backward compatibility and migration

V1 is opt-in. Direct simulation and physical launches retain legacy, PDE off, and observability off where the source defines those defaults. Existing legacy topics, five compatibility messages, arrays, config JSON, wrappers, launches, data readers, and controller/filter owners remain. The robust profile adds typed observability and supervisor behavior around the same owners; it does not require legacy users to migrate runtime code.

Heavy-Ball ESC files under historical paper_recreations and legacy wrapper/ config paths remain useful for reproduction and regression. Their presence in a default filepath does not make Heavy-Ball the active V1 method. Current V1 instructions explicitly select GESG filter/controller configuration and robust_gaussian_v1.

Migration for a new managed V1 run is therefore configuration-only:

1. keep the canonical launch and sole owners;
2. select algorithm_profile:=robust_gaussian_v1, PDE extensions, and typed observability;
3. supply a versioned scenario/metadata contract;
4. use record_run, then validate_run and analyze_run; and
5. retain legacy and failed artifacts unchanged.

The physical snapshot is adapted around hardware, time, rotation, and passive evaluation, while shared algorithm files remain parity-controlled. No simulation-only evaluator topology enters physical control.

17. Reproducibility checklist

Use this checklist for a supported reproduction or audit:

- ☐ Record repository path, branch, full commit, and `git status --short`. For this report the parent is e3dd0ef...; use the full ID returned by the containing-commit command in Section 1 as the V1 closeout boundary.
- ☐ Record Ubuntu/ROS/Gazebo/Python environment and source order. The project was developed against ROS 2 Humble on Ubuntu 22.04-class hosts.
- ☐ Rebuild the reader-facing report from the authoritative source with `tectonic FINAL_PROJECT_REPORT_V1.tex` in docs/codex/gesc_gaussian/, then confirm metadata/page count with `pdftotext` and text extraction with `pdftotext`. The Phase 10 closeout used Tectonic 0.17.0.
- ☐ Build only the required packages and retain bounded build/test output.
- ☐ Confirm algorithm_profile, launch file, cost owner, controller owner, recorder owner, /odom route, cost sign/units, and evaluation-only topics.
- ☐ Copy and complete metadata; never edit a retained run's resolved input.
- ☐ Record scenario schema/version, profile ID, case ID, seed, start, sources, relative intensities, disturbances, exact predicates, timeouts, and run root.
- ☐ Use a new run ID and an empty target directory; do not overwrite or automatically retry failed evidence.
- ☐ Verify readiness before nonzero motion; preserve a failure if it never opens.
- ☐ Stop through the managed recorder and verify readiness false, stop true, stable final zero, bag finalization, integrity, and completeness.
- ☐ Retain sqlite3 bag, metadata/config/parameters, console, completeness, scenario result, analysis summary, CSV/plots, and hashes as applicable.
- ☐ Compare only compatible contracts. Keep formal, behavioral, infrastructure, selected, and broad denominators separate.

- Label unavailable external paths and unexecuted commands; never infer a pass from a handoff that records only a proposal.
- Preserve failures, corrections, and superseded procedures. A corrected new version does not alter the old outcome.
- For physical work, verify backup/rollback, source parity, devices, e-stop/operator clearance, wall clock, passive Vicon identity, manual stop, and no autonomous obstacle avoidance.

The exact Phase 10 report audit is reproducible from [phase_10_report_coverage.tsv](#) and [phase_10_documentation_validation.md](#).

18. Limitations, threats to validity, and unsupported claims

18.1 Algorithm/model limitations

- Counted-candidate V1 assumes two sources and one fillable undesired basin; unknown source count and more than two basins were not established.
- The fitted local quadratic and Gaussian approximation can be poor for asymmetric, flat, overlapping, sparse, or abruptly changing fields.
- Center, width, amplitude, candidate ranking, and escape direction depend on sampling trajectory, timing, dither/rotation phase, windowing, and selected thresholds. V1 showed seed/start sensitivity.
- Open-field assist has no general obstacle/wall planner. Recenter and several post-recovery mechanisms are disabled in the selected terminal profile.
- Sensor noise, delay, contacts, and limited geometries were qualified as infrastructure, not exhaustively sampled as a robustness population.

18.2 Simulation threats

- Gazebo lumen inputs are relative model parameters, not measured lux at the rotating sensor. Shadows, reflections, wall material, occlusion, ambient light, thermal/electrical drift, and physical response were not calibrated into the simulation.
- Many probes were development-selected or corrections after inspecting prior outcomes. M4.8 was explicitly evidence-selected. These cannot estimate an unbiased success probability.
- v8.10, v8.11, and v8.12 changed implementation/evaluator/evidence contracts; their denominators cannot be pooled.
- Only one v8.12 varied matrix case ran; 0/1 `formal` does not mean four empirical failures. Three cases were withheld.
- Optional three-light, arbitrary source count/intensity, broad walls/ obstacles, and the required 70-unique-run suite were not completed.

18.3 Physical threats

- One selected two-light run supplied the strongest physical behavior. It is insufficient for repeatability, population robustness, or transfer claims.
- The physical photoresistor retained volts/-V; no frozen calibrated source-score mapping was used. Later voltage peaks support relative progress, not absolute radiometry.
- Wheel/IMU /odom drifts; Vicon is a different passive evaluation frame and cannot correct control. Frame alignment and coverage can affect comparison.
- The eighth run ended before second convergence/classification/goal; 61/62 completeness includes a known receipt-order false negative that remains uncorrected in V1.
- Open-field operation depended on a human e-stop/clearance and manual stop. No autonomous physical collision avoidance was demonstrated.
- Pi-side run paths in this report come from retained repository evidence; Phase 10 deliberately did not access the currently mounted live Pi home.

18.4 Engineering/evidence limitations

- The inherited broad style suite remained heavily failing (875 at baseline; Phase 09 recorded 917 non-gating style failures, 873 Q000). Focused behavior suites passed at their declared scopes but do not erase that debt.
- Phase 00–07 predated live status/checkpoint files. Handoffs/Git/tests recover the boundary, but the provenance form is less uniform than later phases.
- Raw high-volume run evidence is outside Git and may be unavailable on another machine. Git-side reports, hashes, and manifests are receipts, not replicas.
- Recorder receipt order across independent ROS publishers is not causal order. The eighth shutdown case is the known concrete example.

Accordingly, V1 does **not** support claims of broad simulation robustness, broad physical robustness/readiness, arbitrary-layout/source-count handling, calibrated sim-to-real light equivalence, autonomous obstacle safety, or a formally complete second-extremum physical acceptance run.

19. Future V2 boundary

Future robustness work is intended for a new branch named `feature/gesc-gaussian-robustness-v2`, created **after** the authorized Phase 10 closeout commit is recorded on V1. The new branch must start from the commit containing this report with the exact subject recorded in Section 1, not merely from `parent/base e3dd0ef`. Phase 10 did not create the V2 branch.

Evidence-backed questions suitable for a new Plan include:

- Can a frozen prospective design pass a sufficiently broad unique-run denominator without version-on-first-miss adaptation?
- Which sensing/windowing/fill design changes reduce seed/start sensitivity while preserving cost sign, ownership, and legacy behavior?
- Can candidate classification avoid assuming a known count without using evaluator truth?
- Can escape direction and support geometry handle boundaries, obstacles, overlapping/asymmetric basins, source ratios, reflections, noise, and delay?
- Can simulation inputs be calibrated against measured sensor response with uncertainty, while keeping raw voltage and minimization sign explicit?
- Can the physical workflow reproduce complete local-fill-escape-second- confirmation-GOAL_HOLD behavior across predeclared trials?
- Can the shutdown validator model cross-publisher causal intervals without weakening final-zero safety?
- Which inherited style/package debt should be removed before a new broad gate?

These are study areas, not chosen fixes. V2 requires a new Plan, status, versioned scenarios/parameters, acceptance gates, run roots, validation, checkpoint, handoff, and eventual V2 report. It must preserve V1's report, failed evidence, and denominators unchanged. No V2 branch, code, result, or readiness claim was created in Phase 10.

20. Appendices

Appendix A — implementation-package coverage

The package manifest contains 57 unique, present, Git-tracked path entries: 13 root specifications/navigation documents, 23 prompts, 8 source-material items, 8 templates, and 5 tools. File modes are 52 regular 100644 and five executable 100755. The package has 58 files because MANIFEST.md does not list itself. The manifest is a path inventory dated before later work, not a content-integrity hash.

Every entry has exactly one `package_manifest` row in [phase_10_report_coverage.tsv](#), with authority, outcome status, report section, and role. The groups are:

Package group	Count	Report use
specifications/navigation	13	Requirements, architecture, algorithm, states, data, gates, workflow, topics, risks, physical readiness, navigation.
prompts	23	Durable Phase 00–10 Plan/Implement instructions plus prompt index.
source material	8	Meeting audio/transcripts, consolidated decisions, and two whiteboards.
templates	8	Phase lifecycle, metadata, audit, defaults, scenario, and topic-manifest forms.
tools	5	Status initialization, context/document validation, checkpointing, and context bundling.

Binary/audio/image inputs are summarized and indexed, not reproduced in this Markdown. `templates/robust_gaussian_defaults.yaml` is a design reference; live source/launch and versioned scenario overrides are final parameter authority.

Appendix B — phase and evidence index

Artifact class	Lossless grouping/count at Phase 10 audit	Controlling use
base Phase 00–07 Plans	plans/phase_0[0-7]_plan.md: 8	Original per-phase contracts.
Phase 08 Plans	plans/phase_08*_plan.md: 12	Base/subphase/version contracts.
Phase 09/10 Plans	1 each	Physical and documentation contracts.
Phase 00–07 handoffs	base plus 05.5/07.5: 10	Outcomes and dependency bridges.
Phase 08 handoffs	9	Layered outcomes; 08.8 handoff controls terminal result.
Phase 09 handoff	1 append-only record	Later M8L/eighth amendments supersede its stale header.
statuses	Phase 08, 09, 10: 3	Append-only ledgers/current Phase 10 state.
prior checkpoints	Phase 08 and 09: 2	Material receipts; Phase 10 adds its own closeout checkpoint.
Phase 08 validation	97 tracked files: 77 Markdown, 20 JSON	Complete simulation failure/success/gate history.
Phase 09 validation	21 tracked files	Eight physical-run reports, six snapshot, three Pi, two qualification, two manifests.
Phase 10 validation	coverage TSV and documentation validation	Machine coverage and exact closeout checks.

The coverage TSV also has explicit `phase_result`, `simulation_result`, `physical_milestone`, and `physical_attempt` rows so grouping cannot hide a failed or partial material outcome. Historical status/checkpoint absence is explicit. One historically referenced `docs/codex/gesc_gaussian/phase09_physical_interface_inventory.md` never existed in Git and is not cited as evidence.

Appendix C — interface, topic, and parameter indexes

- Section 8.2 lists all six robust and five compatibility messages.
- Section 8.3 gives canonical runtime/evaluation topic roles.
- Section 9 gives node/direct defaults, selected overrides, units, and constraints for convergence, fill, supervisor, modified cost, controller, recorder, scenario, evaluation, and analysis behavior.
- [Interface map](#) records the final owner graph.
- [Topic dictionary](#) is the field/topic/default reference.
- Current .msg, launch XML, config JSON/YAML, and Python declarations remain authoritative when historical prose differs.

Appendix D — experiment/run/artifact index

Evidence family	Retained location/index	Scope
Phase 05 accepted recording	Phase 05 handoff	First complete 19-topic managed bag/final zero.
Phase 06/07 runs	their handoffs and versioned roots under ~/Experiments/GESC-Gaussian/runs	Scenario/analysis prerequisites.
Phase 08 v1–v6	Section 11.2 and 97-file validation group	Failed/corrected development and gates.
Phase 08.7	Section 11.3	Correction probes and selected reproduction.
Phase 08.8	Section 11.4 and terminal report	v8.0–v8.12, selected successes and broad failure.
Phase 09 snapshots/transfers	Section 12.1–12.2 and Phase 09 validation group	Backup, hash, parity, transfer, rollback, static/check-only.
Physical attempts	Section 12.3 and eight repair/validation reports	No-motion, partial-motion, convergence, and eighth behavioral success.
High-volume products	exact external roots in Sections 11.5, 12.3, and evidence reports	Bags, summaries, CSVs, plots; retained outside Git.

Appendix E — Git chronology

Boundary	Commit(s)
audited baseline	e0c693e
observability	6eaab86, 2ba43f3
supervisor/profile	000f001, 3ebdf70, 571a193, 8cf93f5
robust fill	8717bf5, e48355a, 8789693
escape/recenter	b0e024d, 212c284, 63744b0, 2e40388
recording/bridge	3982eb9 through a7df8cf; bridge 7cc9ace
scenario/analysis/prerequisite	b84518a–8c6191d; 80d5c6a; 2a74001
terminal Phase 08.8	final closeout 9335da2, receipt c04c222
Phase 09 snapshot/physical line	5b4c162 through current e3dd0ef
Phase 10	parent e3dd0ef; closeout commit containing this report with subject <code>docs(phase10): close GESC Gaussian V1 report and evidence</code>

Use `git log --oneline --decorate` and each handoff for the full intermediate sequence. The exact Phase 10 object ID is Git metadata rather than self-embedded text; resolve it with the command in Section 1.

Appendix F — glossary

Term	Meaning
GESC	Gradient extremum-seeking control used by the existing filter/controller path.
PDE/history	Existing position-dependent excitation/history signals used by the algorithm; not a new physical pose source.
Basin/fill	A sampled local minimum and its positive Gaussian cost lift.
Raw cost	Controller minimization input before Gaussian/affine contributions; physical value is $-V$.
Source score	Optional normalized/classification value, distinct from raw cost.
Augmented cost	Weighted sum of raw, Gaussian, and affine components.
Candidate	A convergence event being classified as undesired or goal.
Formal pass	All predicates in one exact frozen contract passed.
Scientific/behavioral pass	Intended behavior occurred even if an evidence/evaluator predicate failed; never automatically formal.
Selected demonstration	Named fixed profile/layout evidence without broad-population inference.
Readiness	Predeclared complete software/behavior gate, not mere launch/static/check-only success.
Final zero	Stable base and, physically, rotation zero observed before recorder teardown.
Vicon	Passive external evaluation odometry; never /odom or controller input.
M8L	Final implemented Phase 09 rotation-initialization repair preceding attempt eight.
V1/V2	Immutable current evidence line and a future separately planned robustness line.

Appendix G — final answer in one paragraph

V1 implemented the requested observable, switchable, adaptive GESC + Gaussian pipeline and demonstrated selected two-basin recovery in simulation and once on the physical robot. It retained substantial negative evidence and did not pass the broad simulation denominator, broad physical readiness, or a complete second-extremum physical acceptance. The complete V1 record is this report, its coverage/validation companions, the immutable phase artifacts, current source, and retained external runs—not any single success ratio in isolation.